

Class A

**MIDTERM EXAM PROJECT REPORT
MODERN PREDICTION AND MACHINE LEARNING**

**Project: Regional Clustering Analysis of Districts/Cities in Central Java
Province**



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1 Introduction

Regional development is an important aspect of data-driven planning because each region may have different educational, economic, and demographic characteristics. Differences in education levels, economic capacity, poverty conditions, and population density can indicate regional disparities that require different policy responses. Therefore, statistical and machine learning approaches can be used to identify regional patterns more systematically.

This project focuses on regional clustering analysis of districts/cities in Central Java Province. Central Java was selected because it consists of 35 districts/cities, which provides sufficient observations for clustering analysis. The data used in this study were obtained from the official publication of Badan Pusat Statistik Provinsi Jawa Tengah, namely Provinsi Jawa Tengah Dalam Angka 2025 (Badan Pusat Statistik Provinsi Jawa Tengah, 2025). Although the publication was released in 2025, the selected indicators refer to 2024 data because they represent the latest complete district/city-level data available in the publication.

The variables used in this study include average years of schooling, GRDP per capita, poverty rate, and population density. These variables represent educational, economic, poverty, and demographic dimensions of regional development. By using these indicators, the analysis aims to identify groups of districts/cities with similar socio-economic characteristics.

The clustering methods applied in this project are Hierarchical Clustering and K-Means Clustering. Both methods are used to compare the consistency and interpretability of the resulting clusters. The clustering results are expected to provide meaningful insights for regional planning and policy recommendations, particularly in identifying developed urban regions and regions that may require further educational and economic intervention.

2 Methods

2.1 Data Source

This study uses secondary data obtained from the official publication of Badan Pusat Statistik Provinsi Jawa Tengah, namely Provinsi Jawa Tengah Dalam Angka 2025 (Badan Pusat Statistik Provinsi Jawa Tengah, 2025). The unit of analysis is districts/cities in Central Java Province, consisting of 29 regencies and 6 municipalities. Therefore, the total number of observations used in this study is 35 districts/cities.

Although the publication was released in 2025, the selected indicators refer to 2024 data because they represent the latest complete district/city-level data available in the publication. The provincial aggregate value was not included in the dataset because the analysis focuses only on district/city-level observations.

2.2 Variable Selection

The variables used in this study were selected based on their relevance to educational, economic, poverty, and demographic conditions. Four numerical variables were used in the clustering analysis, namely average years of schooling, GRDP per capita, poverty rate, and population density.

Average years of schooling was selected as an education indicator because it represents the educational attainment of the population in each district/city. GRDP per capita was selected as an economic indicator because it reflects the relative economic capacity of each region. Poverty rate was included to represent economic vulnerability, while population density was used as a demographic indicator that reflects population concentration in each region.

Table 2.1 Variables Used in the Clustering Analysis

Variable Name	Indicator	Category	Unit	Year
avg_years_schooling	Average Years of Schooling	Education	Year	2024

grdp_per_capita	GRDP per Capita at Current Market Prices	Economic	Thousand Rupiahs	2024
poverty_rate	Percentage of Poor People	Poverty / Social	Percent	2024
population_density	Population Density	Demographic	People per km ²	2024

2.3 Data Preprocessing

Data preprocessing was conducted before applying clustering analysis. The preprocessing steps included checking the dataset structure, verifying data types, checking missing values, identifying duplicate observations, and standardizing numerical variables.

The dataset consisted of 35 observations and 5 columns, including one region name column and four numerical variables. Missing value checking showed that there were no missing values in the selected variables. Duplicate checking was also conducted to ensure that each district/city appeared only once in the dataset.

Since the selected variables have different measurement scales, standardization was applied to all numerical variables before clustering analysis. Average years of schooling is measured in years, GRDP per capita is measured in thousand rupiahs, poverty rate is measured in percentage, and population density is measured in people per square kilometer. Without standardization, variables with larger numerical values could dominate the distance calculation in clustering.

Therefore, z-score standardization was applied using the following formula:

$$Z = \frac{(X - \bar{X})}{s}$$

where X is the original value, $\bar{X}$ is the mean of the variable, s is the standard deviation, and Z is the standardized value.

2.4 Hierarchical Clustering

Hierarchical Clustering was applied to group districts/cities based on similarities in standardized socio-economic indicators. Euclidean distance was used as the distance measure because the dataset consists of numerical variables that had been standardized. Ward's linkage method was selected because it minimizes within-cluster variance and tends to produce compact and interpretable clusters.

A dendrogram was used to visualize the hierarchical grouping structure and to support the determination of the number of clusters. Several numbers of clusters were evaluated using the Silhouette Score to identify the most appropriate cluster solution.

2.5 K-Means Clustering

K-Means Clustering was also applied as a comparison method because it is a commonly used partition-based clustering algorithm for numerical data (James et al., 2021).

The Elbow Method was used to observe the decrease in inertia for different values of k . In addition, the Silhouette Score was calculated to evaluate the separation quality of each cluster solution. The final number of clusters was selected based on both evaluation results and interpretability.

2.6 Evaluation Metrics

The clustering results were evaluated using the Silhouette Score and Davies-Bouldin Index. The Silhouette Score measures how well each observation fits within its assigned cluster compared to other clusters (Rousseeuw, 1987). A higher Silhouette Score indicates better cluster separation.

The Davies-Bouldin Index measures the average similarity between clusters by considering within-cluster compactness and between-cluster separation (Davies & Bouldin, 1979). A lower Davies-Bouldin Index indicates better clustering quality. These metrics were used to compare the performance of Hierarchical Clustering and K-Means Clustering.

3 Discussion

3.1 Case Study

This study focuses on regional clustering analysis of districts/cities in Central Java Province. The case study was selected because Central Java consists of 35 districts/cities, including 29 regencies and 6 municipalities. This number of observations is sufficient for clustering analysis and fulfills the project requirement to use district/city-level observations.

The main objective of this case study is to identify socio-economic and educational patterns across districts/cities in Central Java. The analysis uses four indicators: average years of schooling, GRDP per capita, poverty rate, and population density. These indicators were selected to represent education, economy, poverty, and demographic conditions. By applying clustering analysis, districts/cities with similar regional characteristics can be grouped together and interpreted for regional planning purposes.

The dataset was obtained from the official publication of Badan Pusat Statistik Provinsi Jawa Tengah, namely Provinsi Jawa Tengah Dalam Angka 2025. The selected indicators refer to 2024 data to maintain temporal consistency across variables.

3.2 Work Steps

The analysis was conducted using Google Colab with Python. The work steps in this study were as follows:

1. The dataset was compiled from the BPS publication Provinsi Jawa Tengah Dalam Angka 2025.
2. The observations were arranged at the district/city level, consisting of 35 districts/cities in Central Java Province.
3. Four variables were selected for clustering analysis: average years of schooling, GRDP per capita, poverty rate, and population density.

4. The dataset was checked for missing values, duplicate observations, and incorrect data types.
5. The numerical variables were standardized using z-score standardization to make all variables comparable.
6. Hierarchical Clustering was applied using Euclidean distance and Ward's linkage method.
7. A dendrogram was created to visualize the hierarchical clustering structure.
8. Several numbers of clusters were evaluated using the Silhouette Score.
9. K-Means Clustering was applied using several values of k.
10. The Elbow Method and Silhouette Score were used to determine the optimal number of clusters for K-Means.
11. The final clustering results from Hierarchical Clustering and K-Means Clustering were compared using the Silhouette Score and Davies-Bouldin Index.
12. The clusters were interpreted based on their average socio-economic characteristics.
13. Policy recommendations were formulated based on the regional patterns identified from the clustering results.

The clustering analysis and evaluation metrics were implemented in Python using the scikit-learn library (Pedregosa et al., 2011).

3.3 Results and Discussions

3.3.1 Data Overview

The dataset used in this study consists of 35 observations and 5 columns. The observations represent districts/cities in Central Java Province, while the columns consist of one region name column and four numerical variables. The four numerical variables are average years of schooling, GRDP per capita, poverty rate, and population density.

The data checking process showed that there were no missing values in the selected variables. The duplicate checking process also showed that there were no

duplicate observations. Therefore, all 35 districts/cities were included in the clustering analysis.

The selected variables have different measurement scales. Average years of schooling is measured in years, GRDP per capita is measured in thousand rupiahs, poverty rate is measured in percent, and population density is measured in people per square kilometer. Therefore, z-score standardization was applied before clustering analysis to prevent variables with larger numerical scales from dominating the distance calculation.

Table 3.1 Dataset Structure

Component	Description
Number of observations	35 districts/cities
Number of variables	4 numerical variables
Region	Central Java Province
Year of indicators	2024
Data source	BPS Jawa Tengah Dalam Angka 2025
Missing values	None
Duplicate observations	None
Scaling method	Z-score standardization

3.3.2 Hierarchical Clustering Result

Hierarchical Clustering was conducted using Euclidean distance and Ward's linkage method. The dendrogram was used to observe the structure of similarity among districts/cities. Based on the Silhouette Score evaluation, the best number of clusters for Hierarchical Clustering was two clusters.

The Silhouette Score results for several cluster numbers are presented in Table 3.2

Table 3.2 Silhouette Score of Hierarchical Clustering

Number of Clusters	Silhouette Score
2	0.5898
3	0.4041
4	0.3929

5	0.4006
6	0.3714

The highest Silhouette Score was obtained when the number of clusters was two, with a value of 0.5898. This indicates that the two-cluster solution provides the best separation among the tested alternatives. Therefore, the final Hierarchical Clustering result was determined using two clusters.

The first cluster consists of Kudus and several municipalities, namely Kota Magelang, Kota Surakarta, Kota Salatiga, Kota Semarang, Kota Pekalongan, and Kota Tegal. This cluster has higher average years of schooling, higher GRDP per capita, lower poverty rate, and higher population density. Therefore, this cluster can be interpreted as a High Education–High Income Urban Region.

The second cluster consists of the remaining districts in Central Java. This cluster has lower average years of schooling, lower GRDP per capita, higher poverty rate, and lower population density compared to the first cluster. Therefore, this cluster can be interpreted as a Developing and Economically Vulnerable Region.

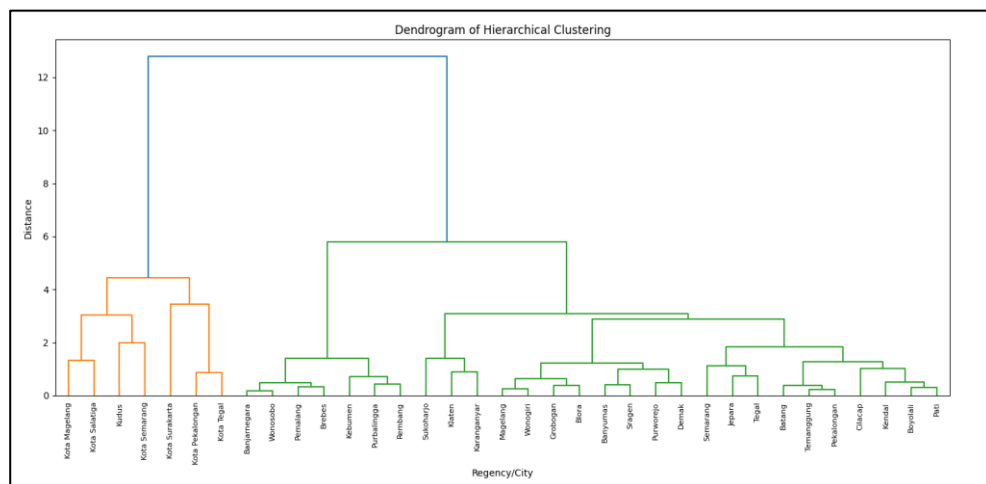


Figure 3.1 Dendrogram of Hierarchical Clustering

3.3.3 K-Means Clustering Result

K-Means Clustering was applied as a comparison method. Several values of k were tested to identify the most appropriate number of clusters. The Elbow Method was used to observe the decrease in inertia, while the Silhouette Score was used to evaluate cluster separation.

The Silhouette Score results for K-Means Clustering are presented in Table 3.3

Table 3.3 Silhouette Score of K-Means Clustering

Number of Clusters	Silhouette Score
2	0.5898
3	0.4041
4	0.3929
5	0.4006
6	0.3331

The highest Silhouette Score for K-Means Clustering was obtained when $k=2$, with a value of 0.5898. This result indicates that the two-cluster solution provides the best separation among the tested values of k . Therefore, the final K-Means Clustering model was also formed using two clusters.

The K-Means result produced the same substantive grouping pattern as the Hierarchical Clustering result. One group represents districts/cities with higher education, higher economic capacity, lower poverty, and higher population density. The other group represents districts/cities with lower education, lower GRDP per capita, higher poverty, and lower population density.

Although the cluster numbers may differ between Hierarchical Clustering and K-Means Clustering, the interpretation remains the same. Cluster numbers are technical labels, while the substantive meaning of the clusters is determined based on their characteristics.

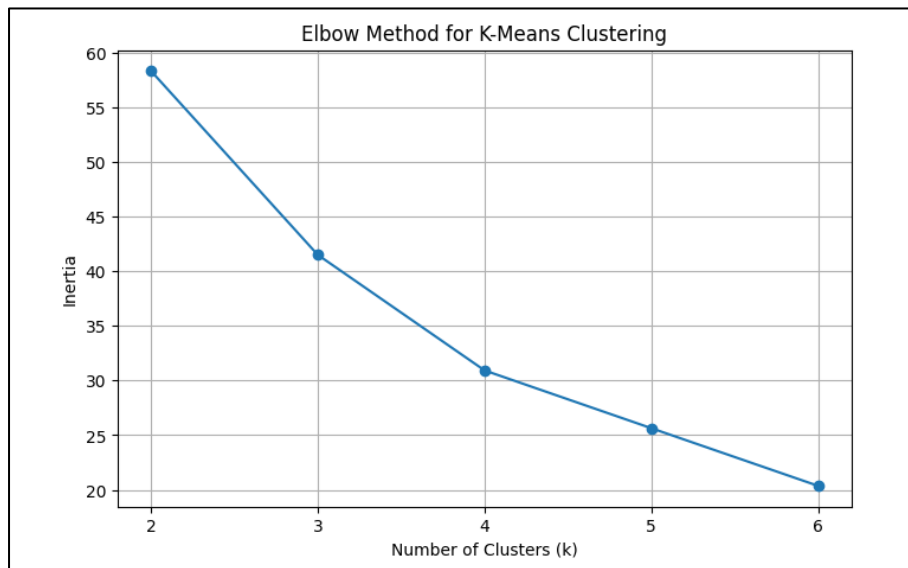


Figure 3.2 Elbow Method for K-Means Clustering

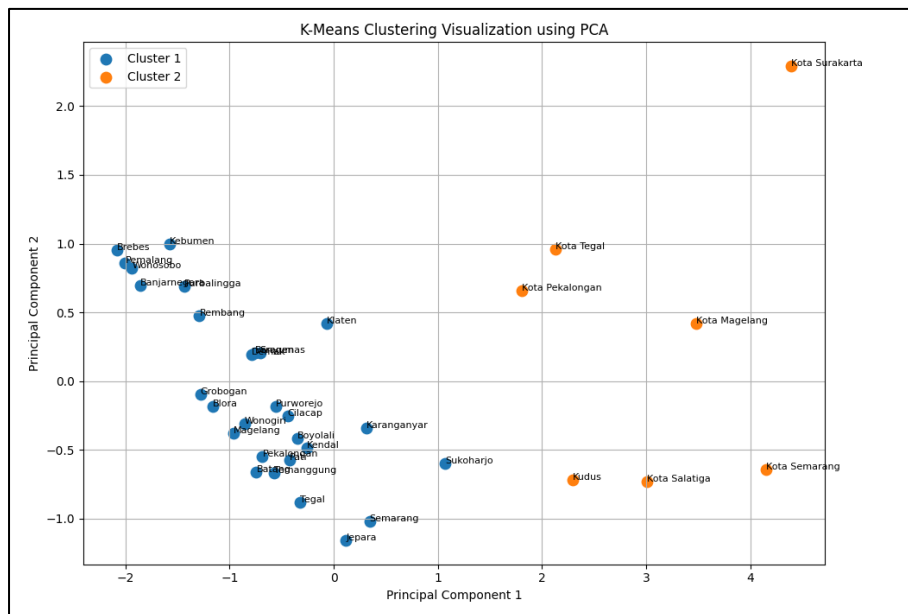


Figure 3.3 K-Means Clustering Visualization using PCA

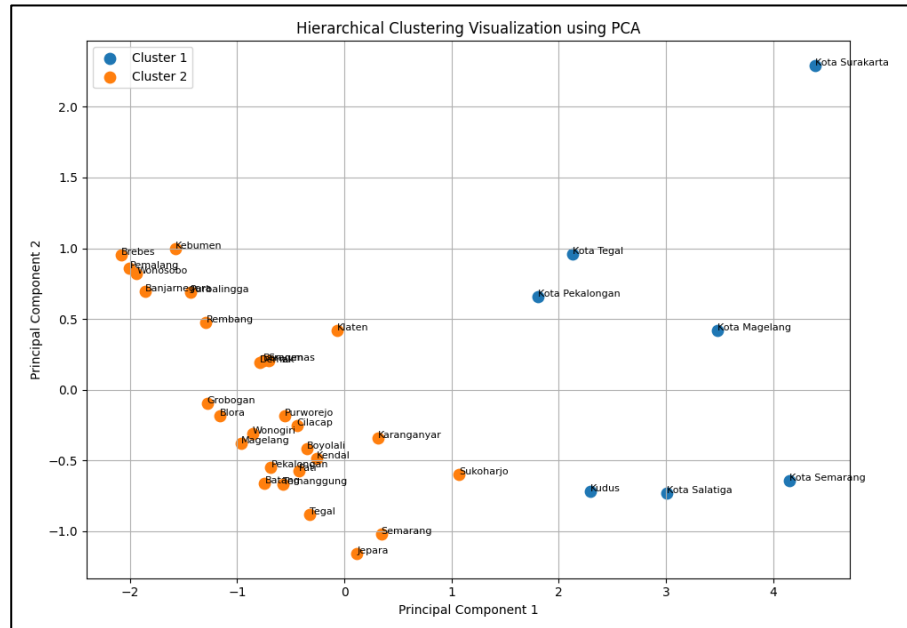


Figure 3.4 Hierarchical Clustering Visualization using PCA

3.3.4 Cluster Interpretation

The final clustering results indicate that the districts/cities in Central Java can be grouped into two major regional development patterns. The interpretation of each cluster was based on the average values of average years of schooling, GRDP per capita, poverty rate, and population density.

The first substantive cluster can be labeled as High Education–High Income Urban Region. This cluster consists of Kudus and several municipalities, namely Kota Magelang, Kota Surakarta, Kota Salatiga, Kota Semarang, Kota Pekalongan, and Kota Tegal. This group is characterized by higher average years of schooling, higher GRDP per capita, lower poverty rate, and higher population density. These characteristics indicate that the regions in this cluster tend to have stronger educational and economic conditions compared to other regions.

The presence of cities in this cluster is reasonable because urban areas generally have better access to education, more diverse economic activities, and denser population concentration. Kudus also belongs to this group because it has a relatively high GRDP per capita, which indicates strong regional economic capacity.

The second substantive cluster can be labeled as Developing and Economically Vulnerable Region. This cluster consists of the remaining districts in Central Java. Compared to the first cluster, this group has lower average years of schooling, lower GRDP per capita, higher poverty rate, and lower population density. These characteristics indicate that the regions in this cluster may still face challenges related to educational attainment, economic development, and poverty reduction.

The cluster interpretation shows that there is a clear socio-economic distinction between urban and more developed regions and the majority of districts that still require further development support. Therefore, the clustering results are not only statistical outputs but also provide meaningful insights for regional planning.

Table 3.4 Interpretation of Final Clusters

Cluster Label	Main Characteristics	Regional Meaning
High Education–High Income Urban Region	Higher education, higher GRDP per capita, lower poverty, higher density	More developed urban/economic regions
Developing and Economically Vulnerable Region	Lower education, lower GRDP per capita, higher poverty, lower density	Regions requiring education and economic development support

3.3.5 Comparative Analysis

The clustering results from Hierarchical Clustering and K-Means Clustering were compared based on cluster separation quality, interpretability, consistency of grouping, and practical usefulness for regional planning.

Based on the evaluation metrics, both methods produced identical results. Hierarchical Clustering obtained a Silhouette Score of 0.5898 and a Davies-Bouldin Index of 0.7242. K-Means Clustering also obtained a Silhouette Score of 0.5898 and a Davies-Bouldin Index of 0.7242. Since a higher Silhouette Score indicates better cluster separation and a lower Davies-Bouldin Index indicates better clustering quality, both methods can be considered to have the same performance in this study.

The identical evaluation results indicate that the two-cluster structure is stable and consistent across different clustering approaches. This means that the socio-economic differences among districts/cities in Central Java are strong enough to be detected by both hierarchical and partition-based clustering methods.

In terms of interpretability, Hierarchical Clustering provides an advantage because the dendrogram shows the structure of similarity among regions. The dendrogram helps explain how districts/cities are gradually grouped based on their standardized socio-economic characteristics. Meanwhile, K-Means Clustering provides a simpler and more direct partition of regions into two groups.

In terms of practical usefulness, both methods are useful for regional planning because they produce the same substantive regional patterns. The results clearly distinguish between a more developed urban cluster and a developing or economically vulnerable cluster. This distinction can support policy makers in identifying regions that require different development priorities.

Overall, there is no clear quantitative superiority between Hierarchical Clustering and K-Means Clustering in this study. However, Hierarchical Clustering is more useful for explaining the grouping process visually, while K-Means Clustering is useful as a confirmation that the two-cluster structure is consistent.

Table 3.5 Comparison of Hierarchical Clustering and K-Means Clustering

Evaluation Aspect	Hierarchical Clustering	K-Means Clustering
Optimal number of clusters	2	2
Silhouette Score	0.5898	0.5898
Davies-Bouldin Index	0.7242	0.7242
Cluster separation	Reasonably good	Reasonably good
Interpretability	Strong, supported by Dendrogram	Good, simple partition
Practical usefulness	Useful for explaining regional structure	Useful for confirming stable grouping

Based on the clustering results, two major regional patterns emerge in Central Java. The first pattern represents urban and economically stronger regions with higher educational attainment and lower poverty. The second pattern represents developing regions that still require stronger policy attention, particularly in education improvement, poverty reduction, and local economic development. These findings show that clustering analysis can transform raw regional indicators into meaningful information for regional planning and policy formulation.

4 Conclusion and Policy Recommendation

This study applied Hierarchical Clustering and K-Means Clustering to identify socio-economic and educational patterns across districts/cities in Central Java Province. The analysis used four indicators, namely average years of schooling, GRDP per capita, poverty rate, and population density. All indicators referred to 2024 data obtained from the publication Provinsi Jawa Tengah Dalam Angka 2025. The clustering results show that the optimal number of clusters is two. Both Hierarchical Clustering and K-Means Clustering produced the same evaluation results, with a Silhouette Score of 0.5898 and a Davies-Bouldin Index of 0.7242. These results indicate that the two-cluster solution provides a reasonably good separation and produces relatively compact clusters.

The first cluster can be interpreted as a High Education–High Income Urban Region. This cluster consists of Kudus and several municipalities, namely Kota Magelang, Kota Surakarta, Kota Salatiga, Kota Semarang, Kota Pekalongan, and Kota Tegal. These regions are characterized by higher educational attainment, higher GRDP per capita, lower poverty rate, and higher population density. This pattern indicates that urban regions and economically strong areas tend to have better socio-economic conditions.

The second cluster can be interpreted as a Developing and Economically Vulnerable Region. This cluster consists of the remaining districts in Central Java. These regions generally have lower average years of schooling, lower GRDP per capita, higher poverty rate, and lower population density. This pattern indicates that many districts still require policy attention, particularly in education improvement, poverty reduction, and local economic development.

Based on these findings, several policy recommendations can be proposed. First, regions in the Developing and Economically Vulnerable Region cluster should receive priority in educational development programs, such as improving

school accessibility, increasing educational participation, and supporting scholarship allocation. Second, poverty reduction programs should be strengthened in districts with relatively high poverty rates through targeted social assistance, community empowerment, and employment programs. Third, local economic development should be supported by improving infrastructure, strengthening micro, small, and medium enterprises, and encouraging investment in less developed districts. Fourth, urban regions in the High Education–High Income Urban Region cluster should focus on maintaining education quality, managing urban density, and improving inclusive economic opportunities.

Overall, the clustering analysis shows that regional development in Central Java is not uniform. There is a clear distinction between more developed urban/economic regions and districts that still face socio-economic vulnerability. Therefore, data-driven clustering analysis can support regional planning by helping policy makers identify regional disparities and design more targeted development policies.

5 References

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6 Appendix

The complete Python code, dataset, standardized dataset, and final clustering results are provided as supplementary files.

1. Google Colab Notebook:

UTS_MPML_Regional_Clustering_Central_Java.ipynb

2. Cleaned Dataset:

01_cleaned_central_java_clustering_2024.csv

3. Standardized Dataset:

02_standardized_central_java_clustering_2024.csv

4. Hierarchical Clustering Dataset:

03_hierarchical_clustering_result.csv

5. Final Clustering Results:

04_final_clustering_results_central_java_2024.csv

6. Data Source:

Badan Pusat Statistik Provinsi Jawa Tengah, Provinsi Jawa Tengah Dalam Angka 2025.